

Motivation: Why HPO?

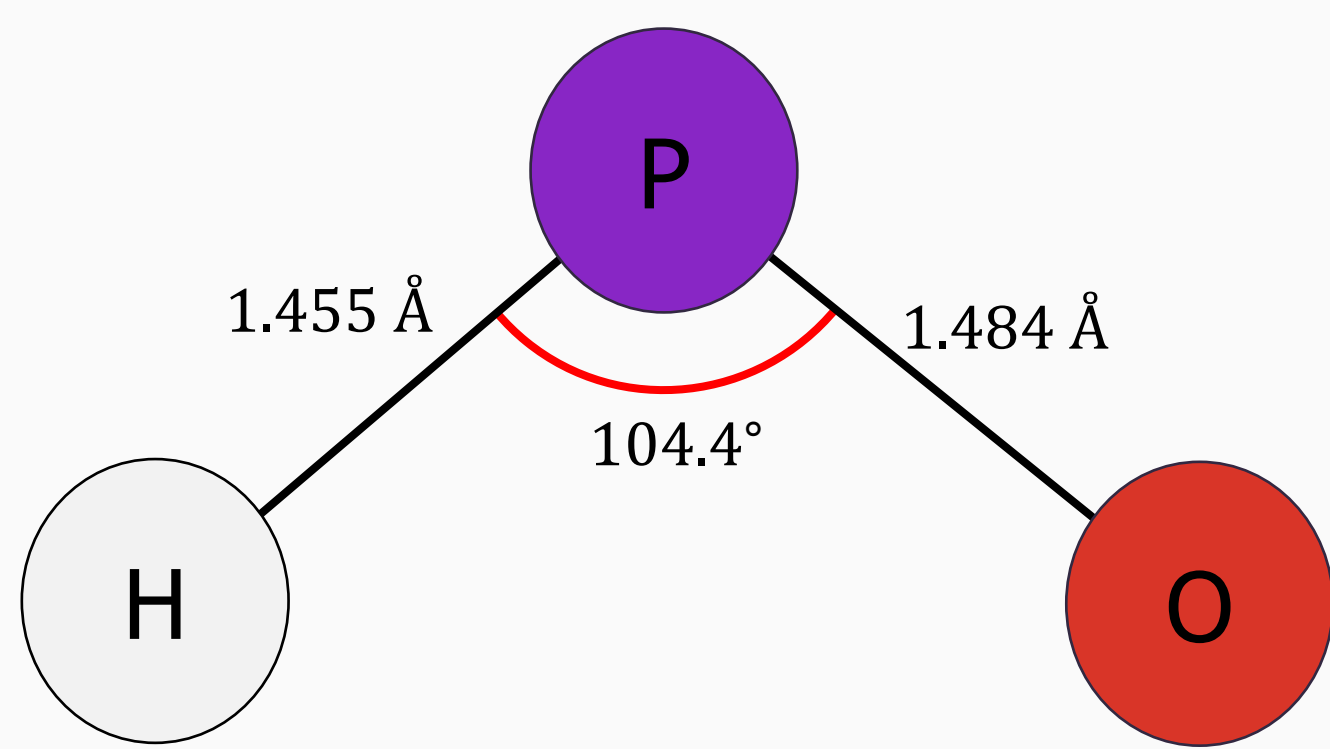
HPO appears as a precursor molecule in proposed abiotic mechanisms to form phosphine on Venus. It was predicted to be detectable in the interstellar medium.

Can HPO play a role on lava or rocky exoplanets?

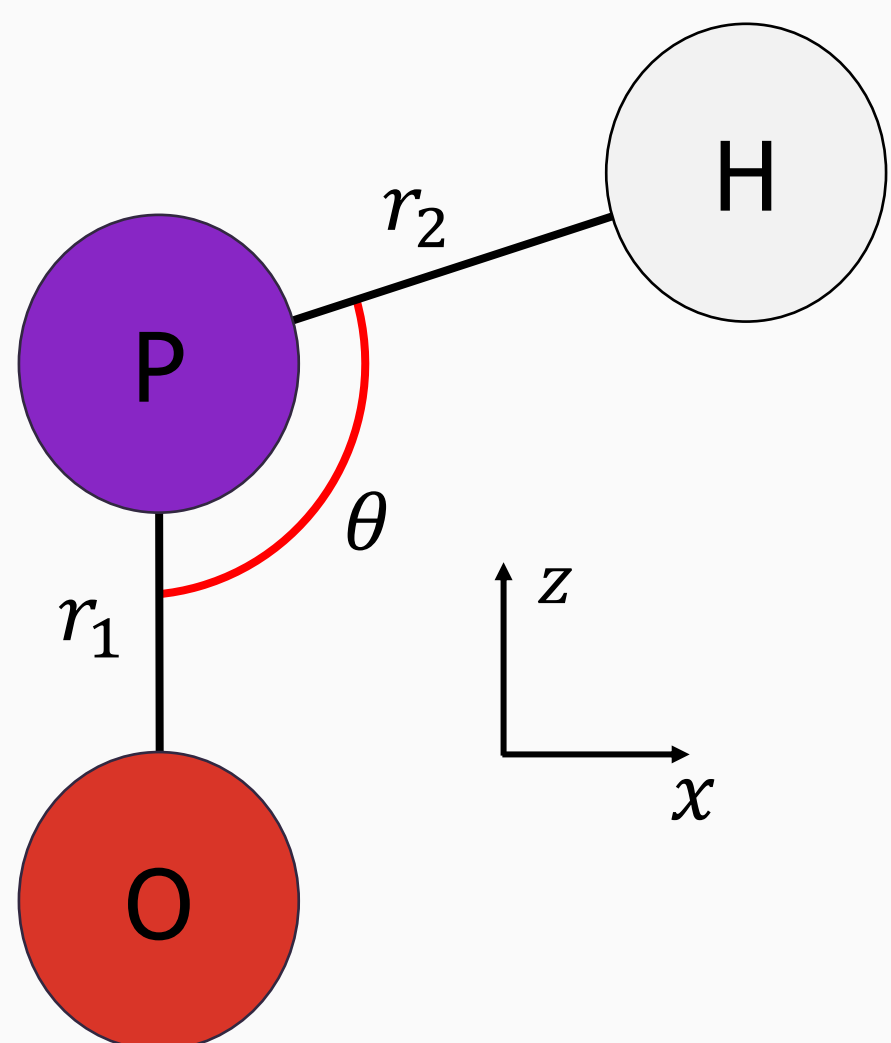
Aims

- Produce a rovibrational linelist for HPO (electronic ground state only).
- Evaluate potential for HPO detection on exoplanets/planets.

Equilibrium bond lengths and angles at the electronic ground state:

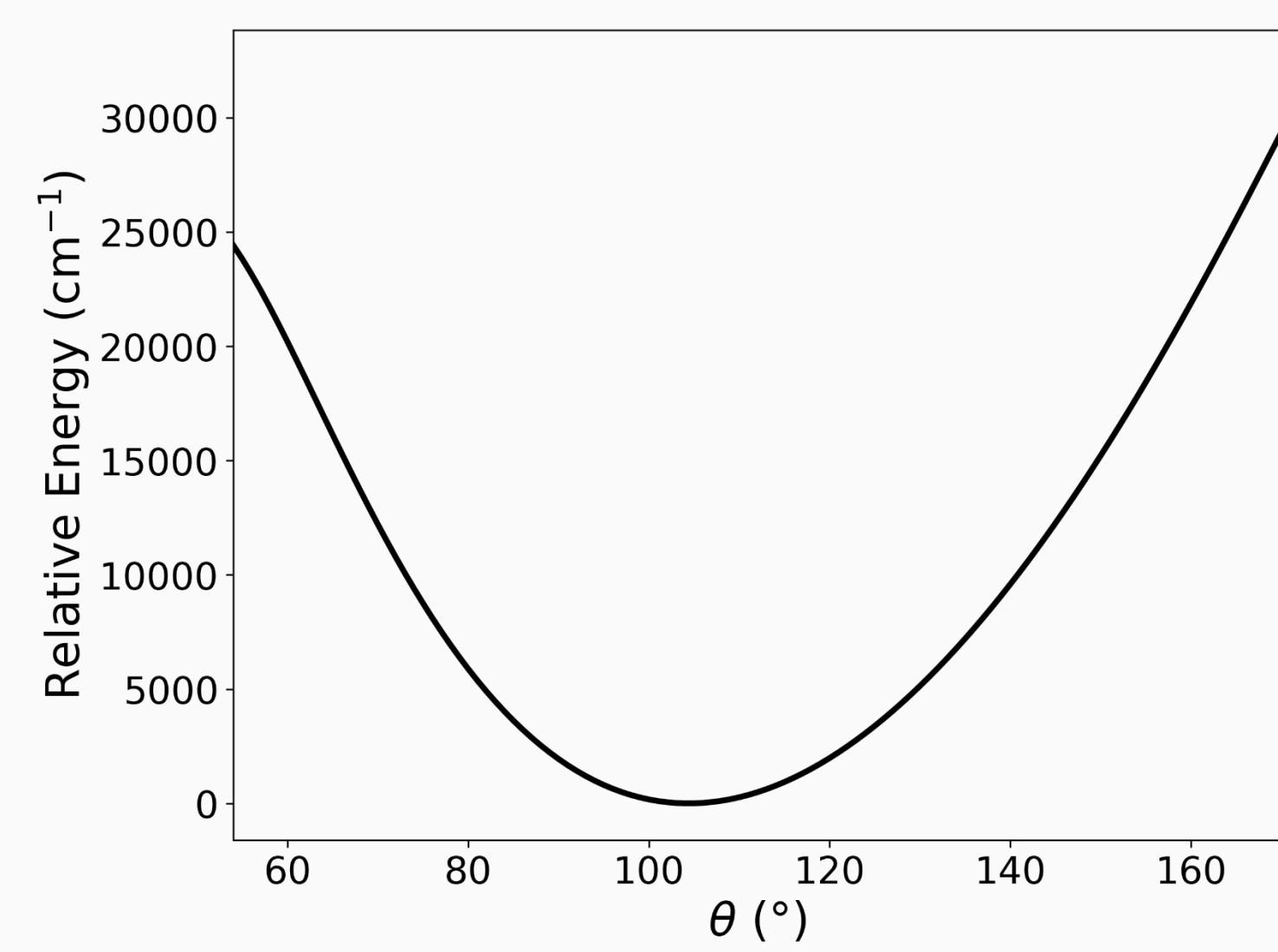
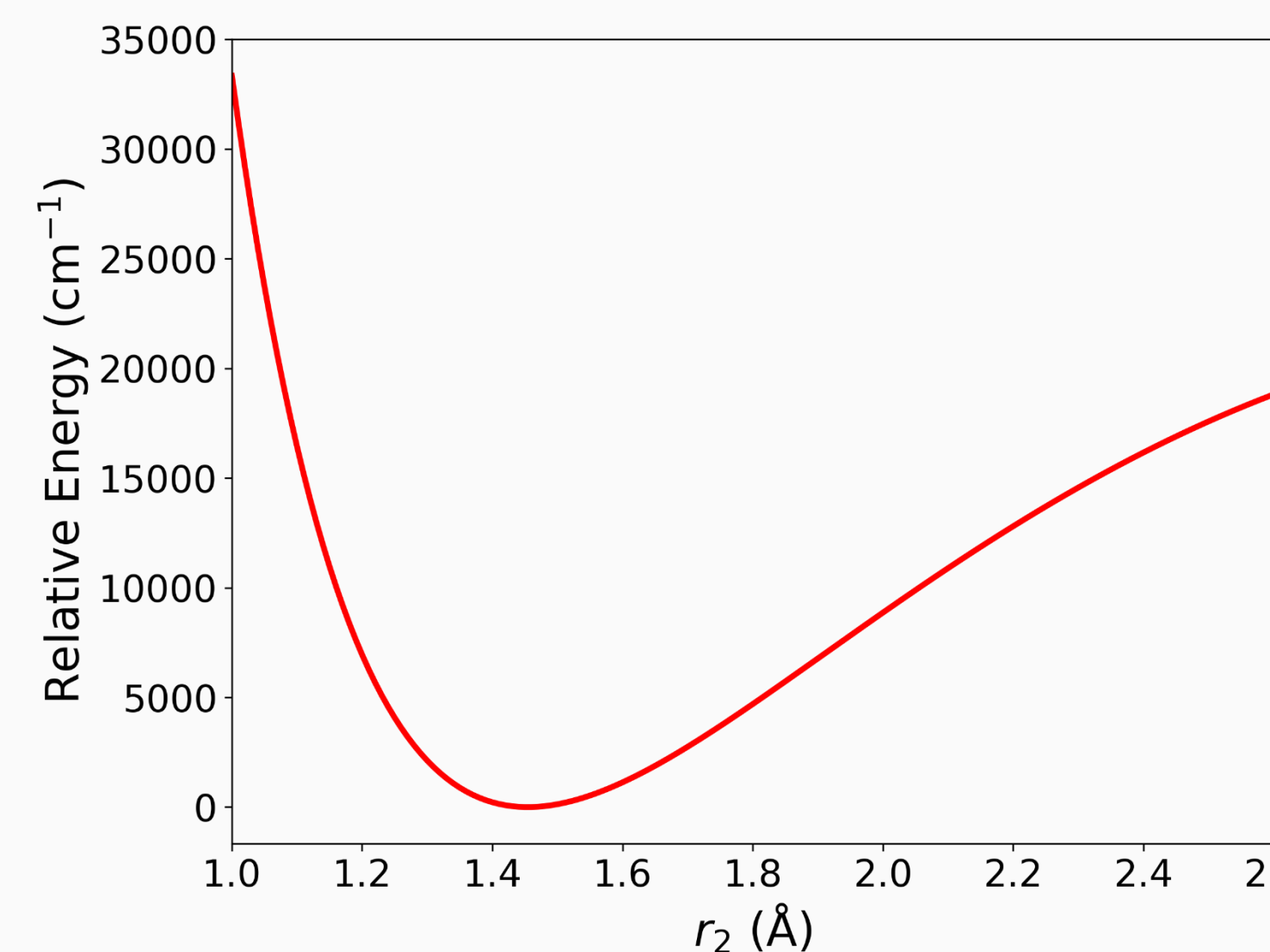
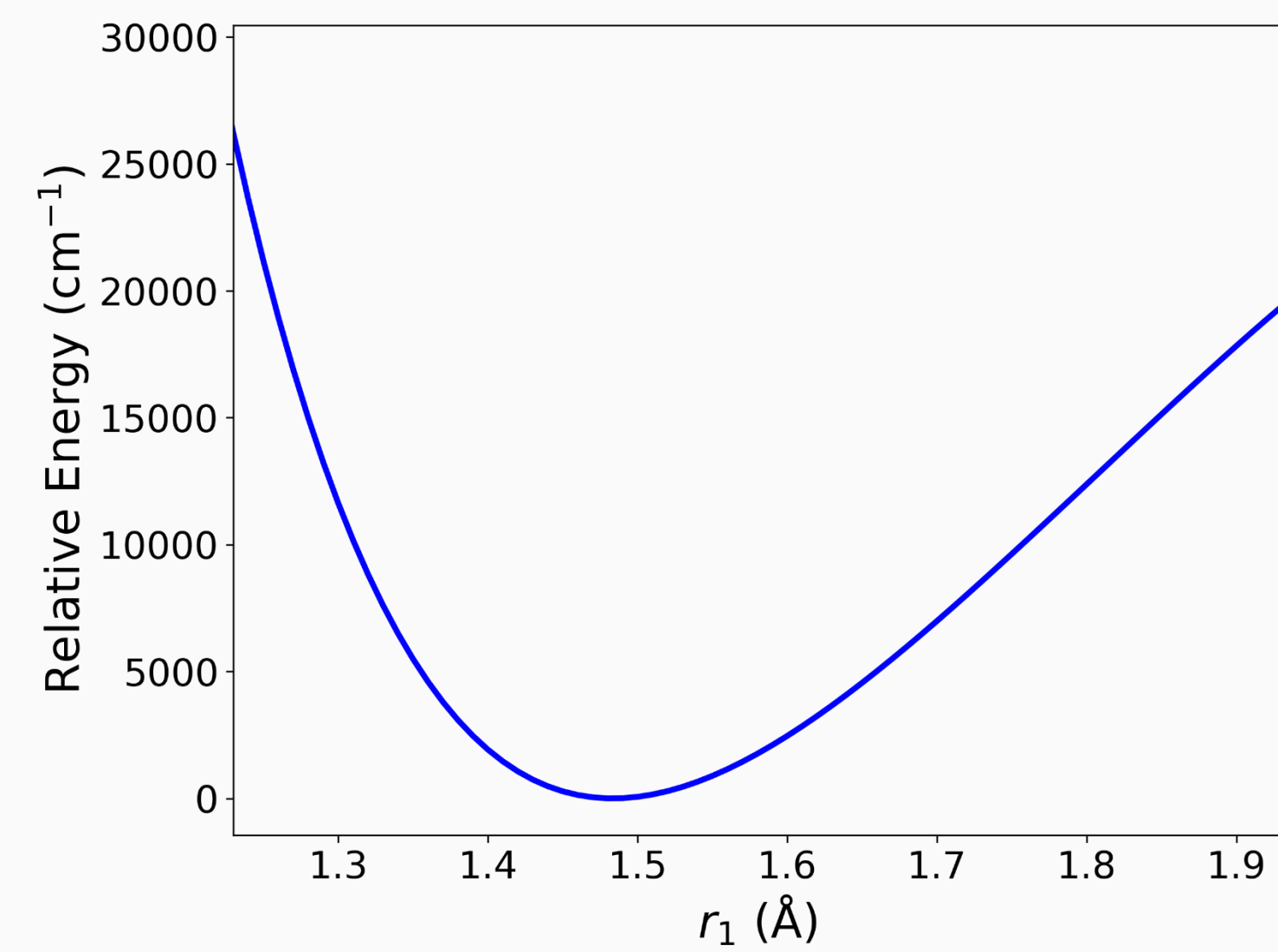


We use the internal coordinates and Cartesian axis as presented:



Potential Energy Surface

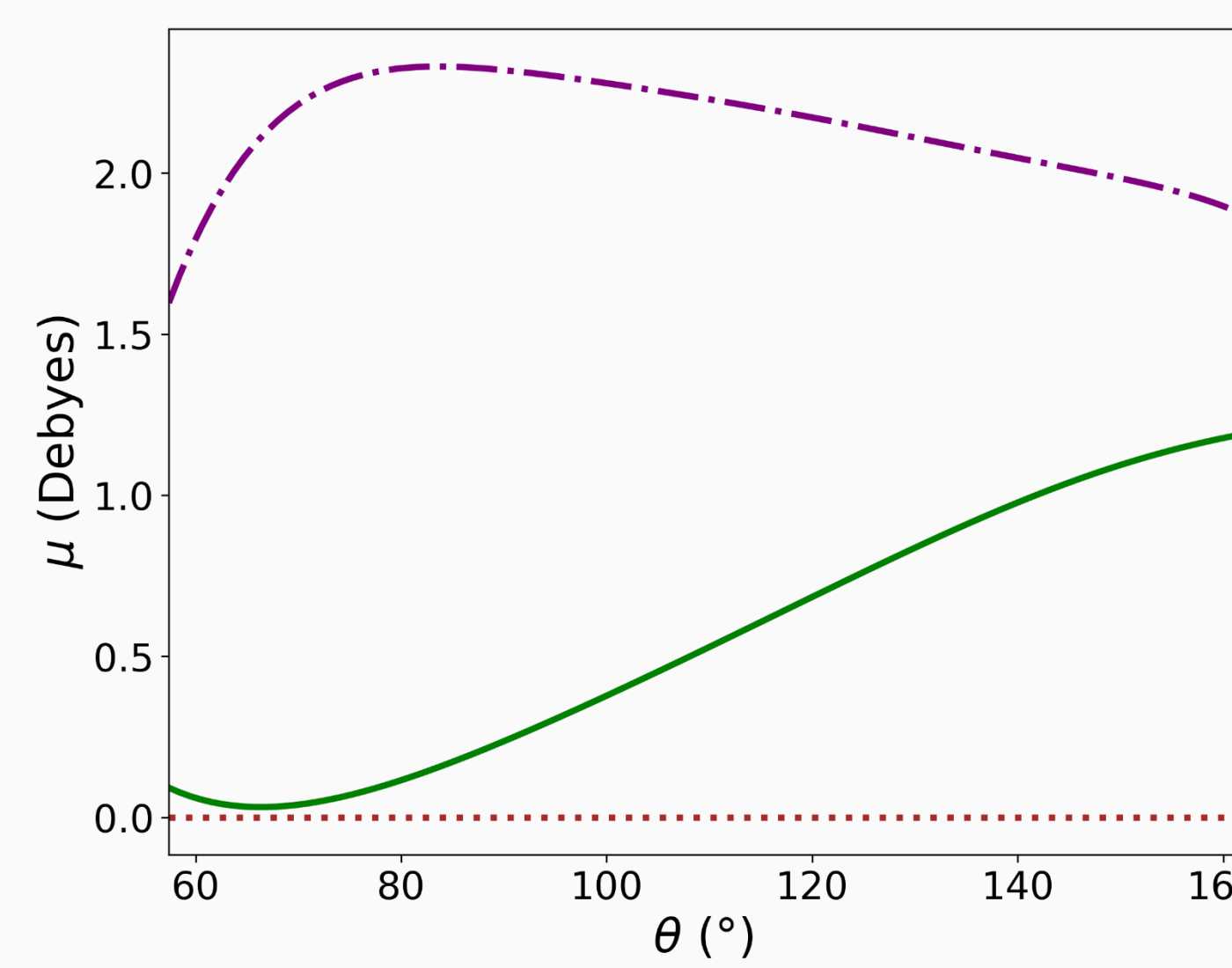
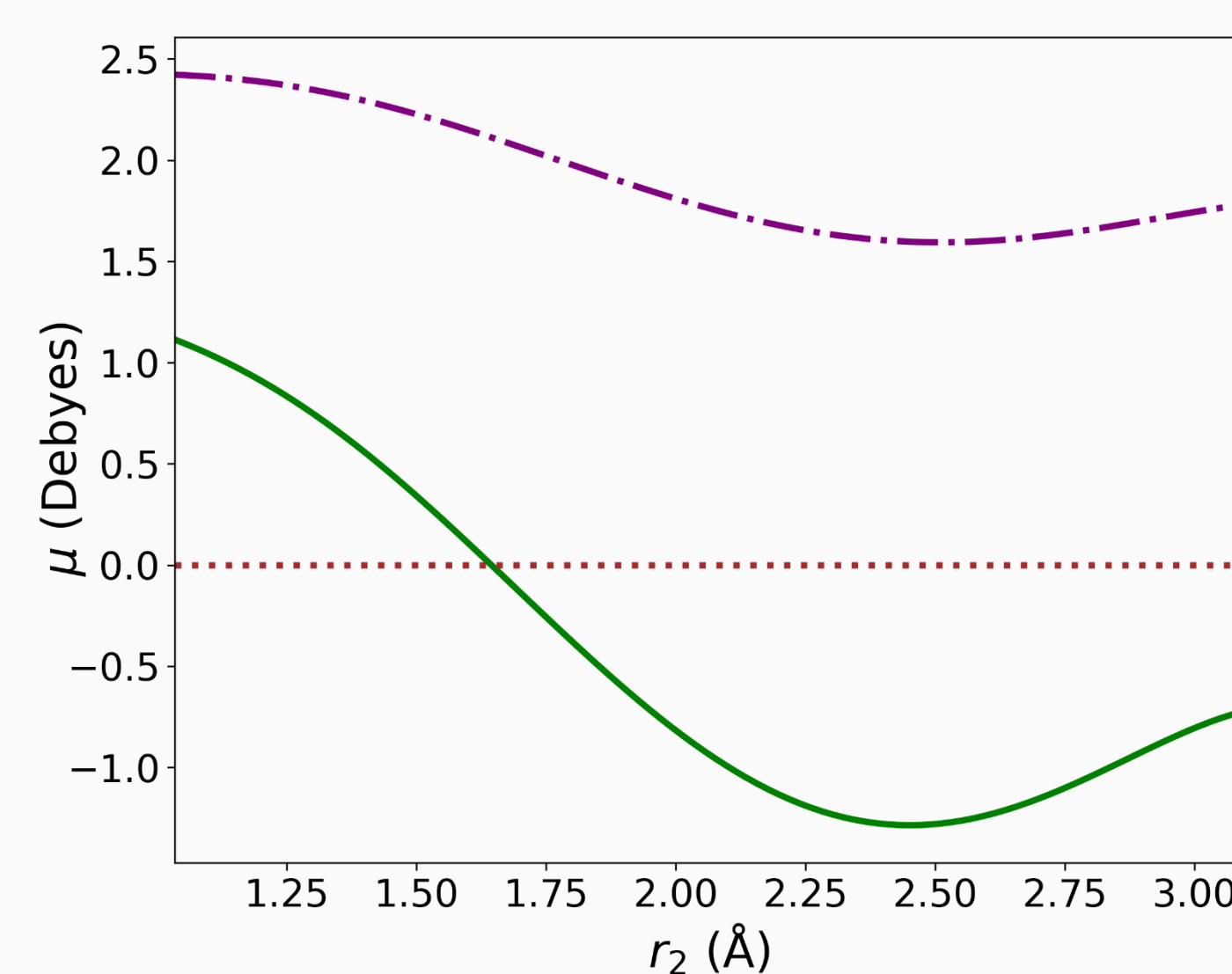
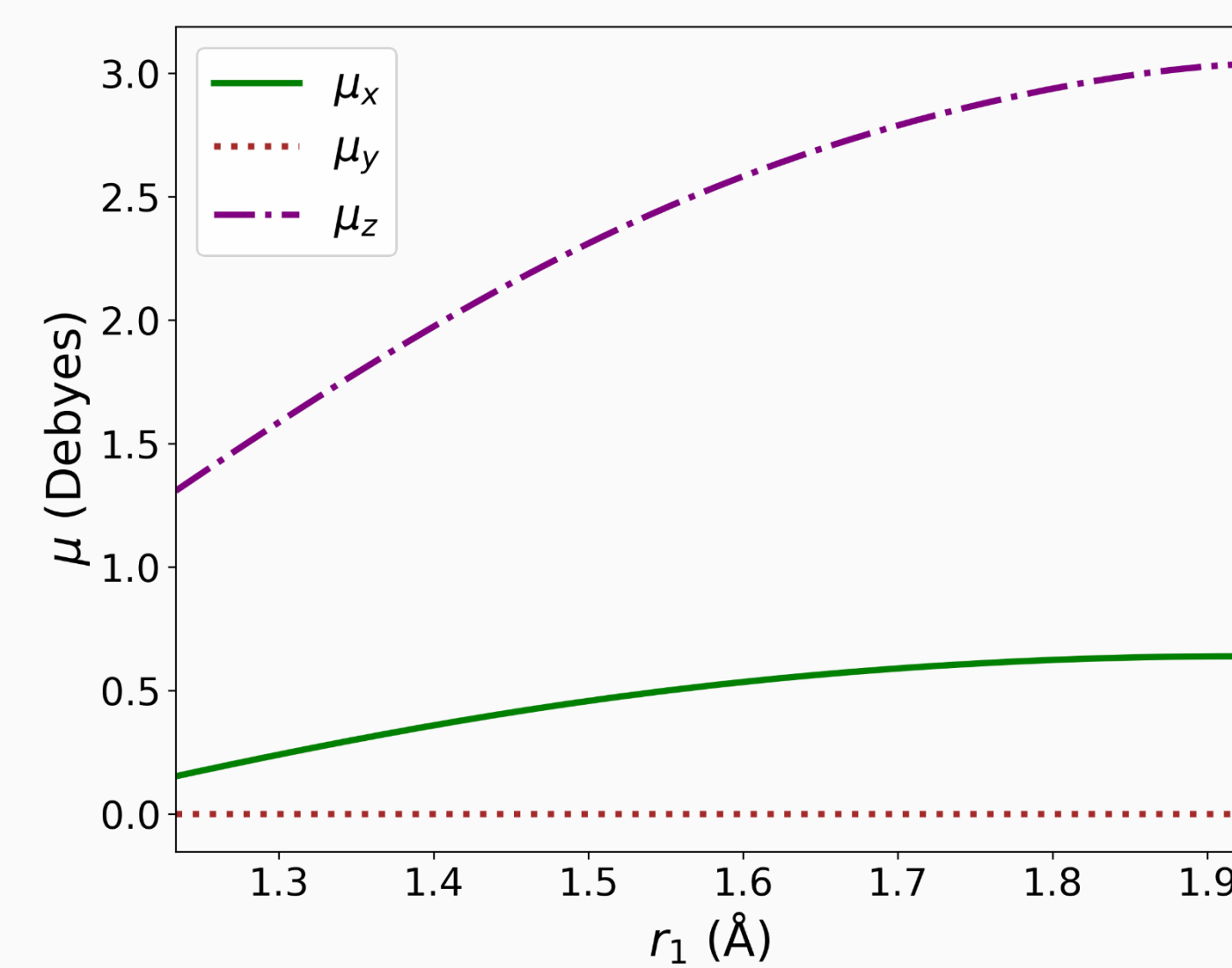
7770 *ab initio* points computed with CCSD(T)-F12/cc-pVQZ-F12a



- Fit PES has root-mean-square error of 2 cm⁻¹ for energies under 18,000 cm⁻¹.
- Molecule appears relatively rigid (stiff bending), we can use non-singular methods to perform variational calculations.

Dipole Moment Surfaces

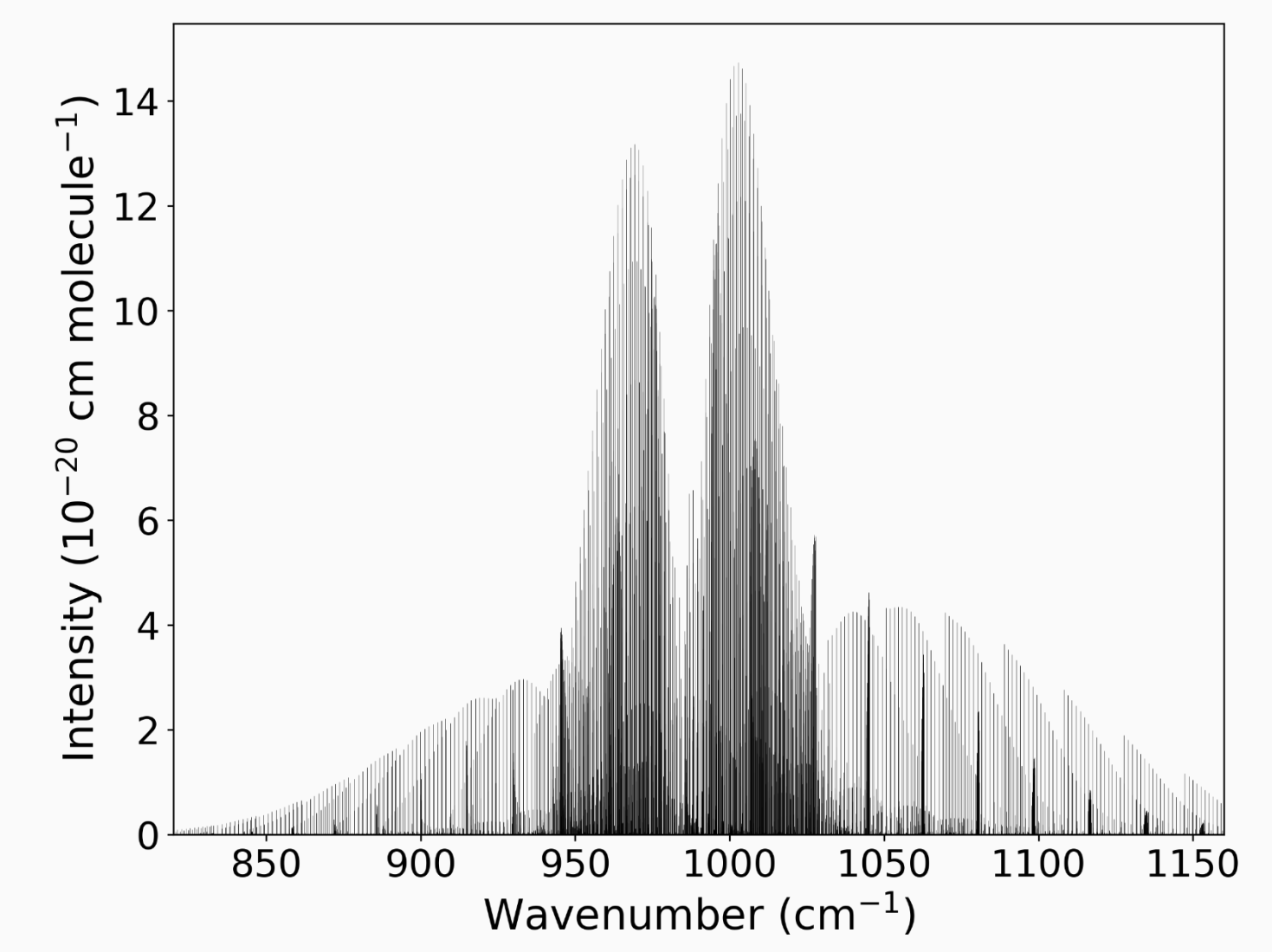
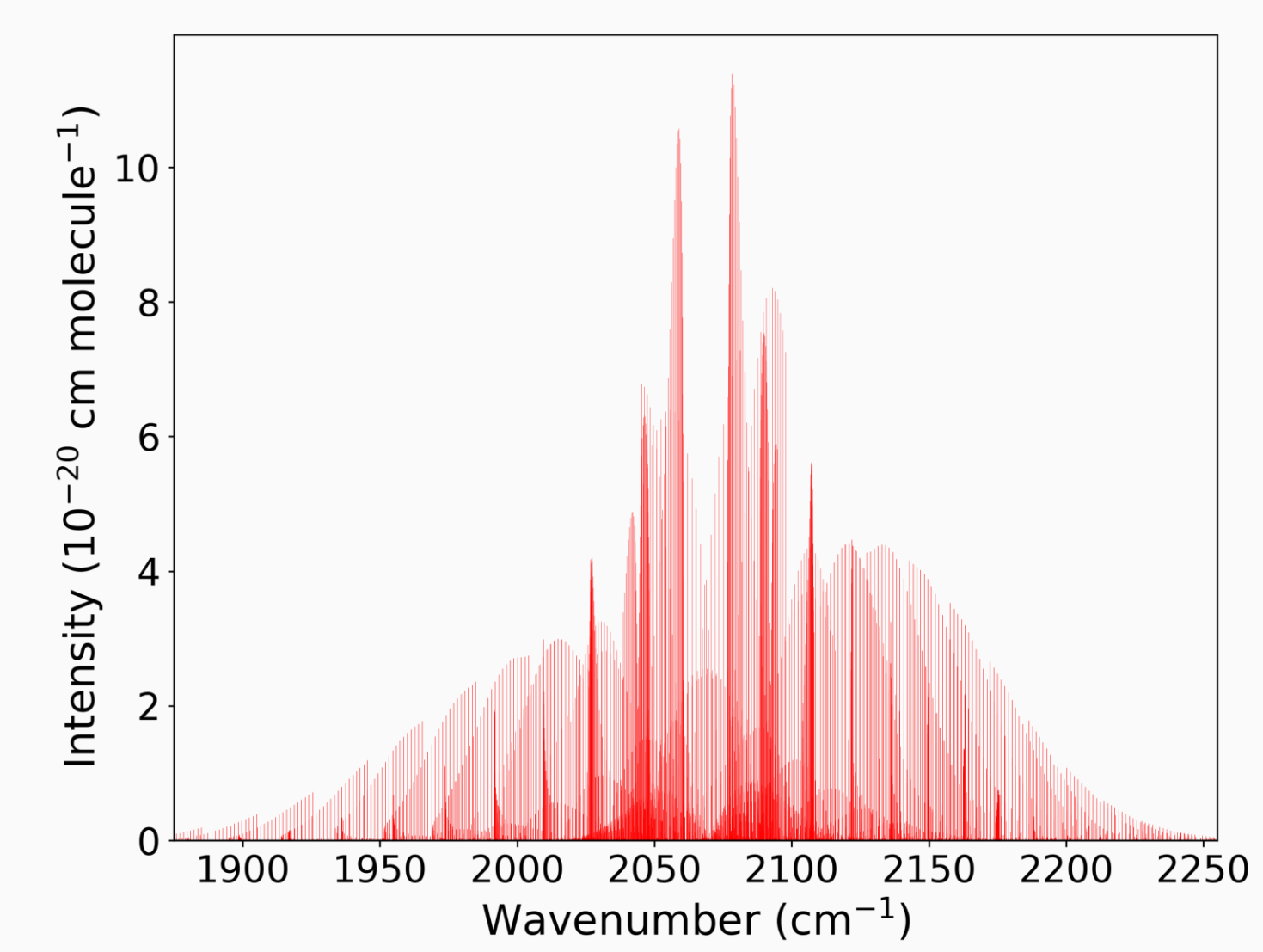
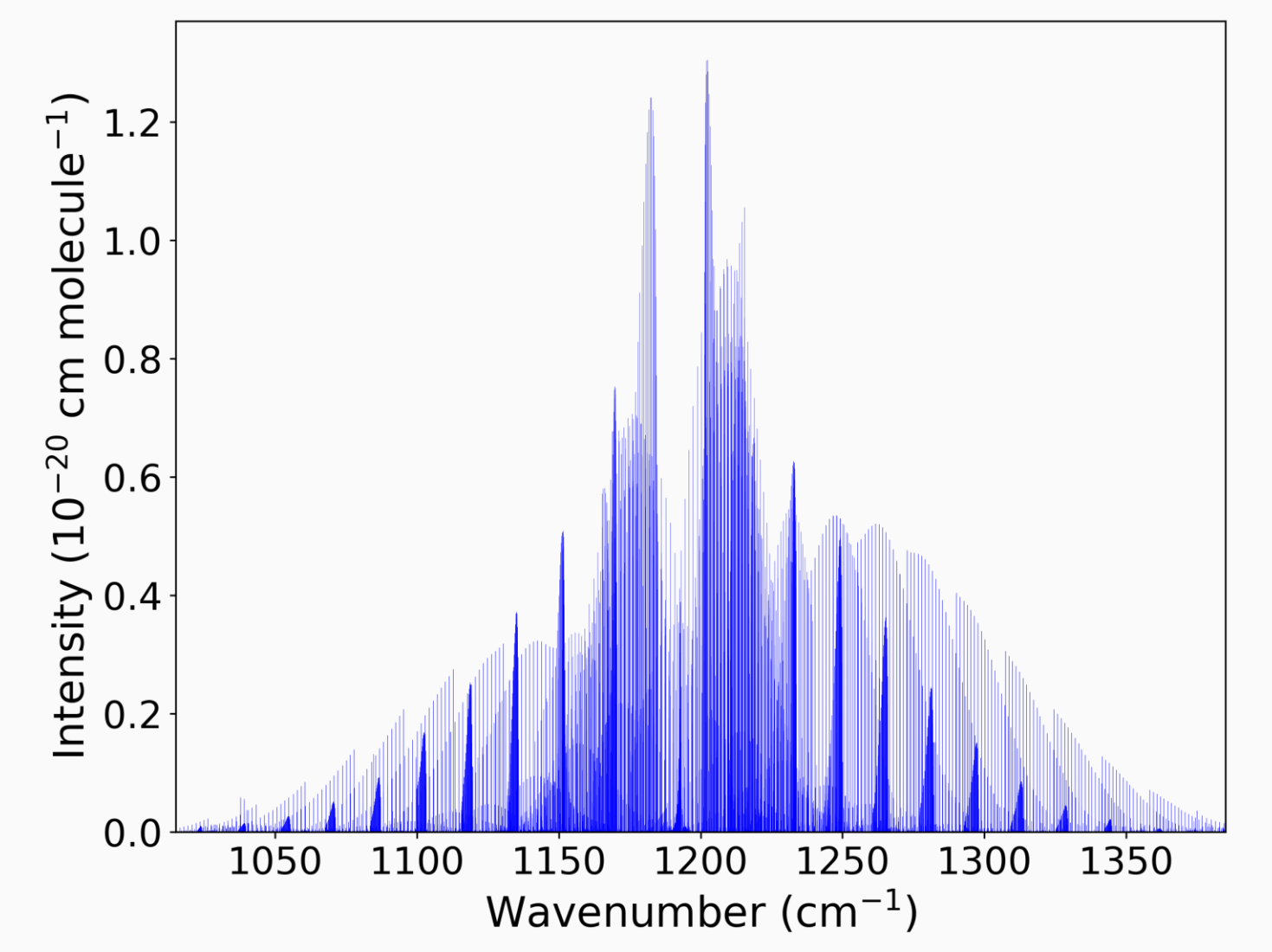
Same grid as the PES computed with CCSD(T)/aug-cc-pV(5+d)Z



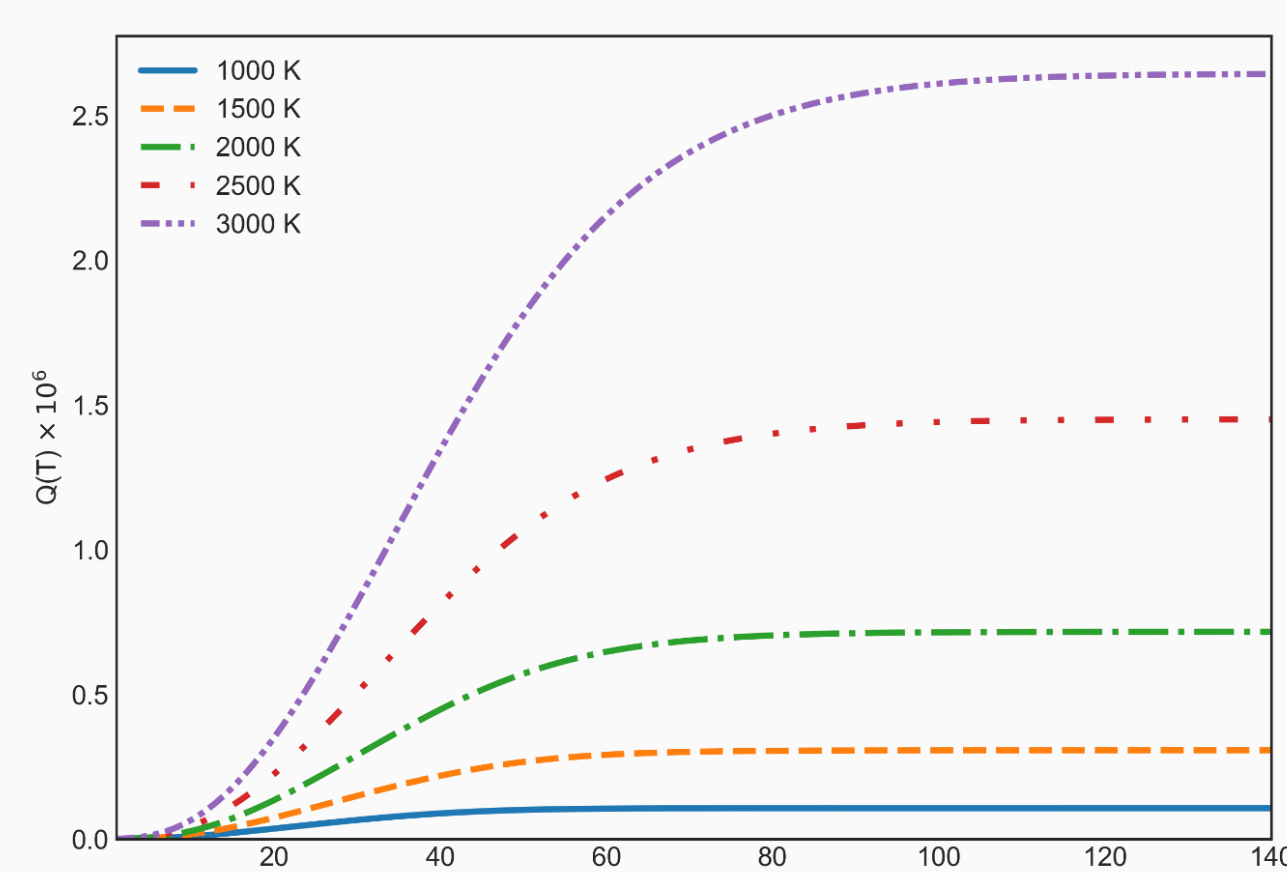
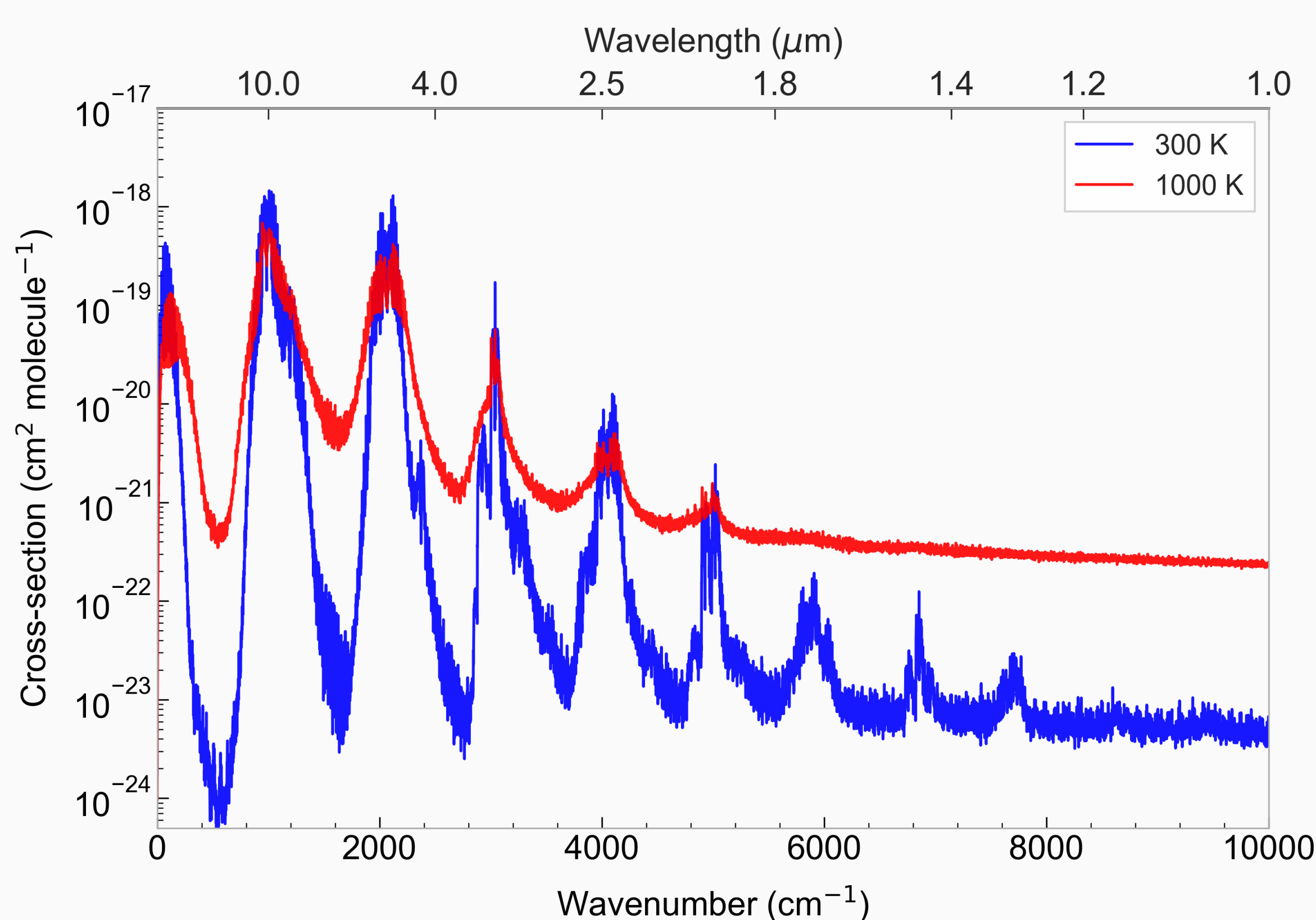
- Dipole values calculated oriented with respect to the Cartesian axis given.
- Fitted to a Taylor-series type expansion.

Stick Spectra

Rotational stick spectra of the fundamental vibrational modes



- The first and third fundamental modes have been experimentally measured, but not the second, despite intensity seeming higher.



Cross-sections computed with states up to $J = 141$. Partition function plots as a function of J suggests $J = 120$ could be sufficient instead.

Cross-sections and partition functions were computed with the program ExoCross.

Cross-sections

Rovibrational energies and wavefunctions are computed with variational nuclear motion program TROVE, then used to compute stick spectra and cross sections.

Lower states are selected between 0 to 14,000 cm⁻¹ while upper states are between 0 to 24,000 cm⁻¹. Includes 2.4 million rovibrational states and 2.2×10^{10} transitions.

Exoplanet Detection?

The fundamental bands of HPO do not overlap with water. This makes detection more feasible. The fundamental band at 985 cm⁻¹ would be particularly visible.

We suggest HPO can be of interest for studying exoplanets and the planetary chemistry of phosphorus.

